Research Square

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# Identification and Classification of Skin Diseases using Deep Learning Techniques

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# Abstract

Health is the most important aspect of a human's life. Humans are covered by the largest organ of their body, the skin. Protecting the body from germs and regulating body temperature are the basic functionalities of the skin. If the skin is not protected, then it may lead to skin diseases caused by bacterial infections, fungal infections, etc. The cost involved in diagnosing skin diseases may not seem cost-effective to everyone. Moreover, dermatologists need to use their professional experience and may have to invest more time in diagnosing skin diseases which usually involves the use of images captured using a dermoscope. The proposed methodology adopts Deep Learning techniques for identifying and classifying skin diseases caused by bacteria and fungi by making use of non-dermoscopic images. The problem statement is an image classification task. The dataset consists of five classes which include two classes of bacterial skin disease namely Cellulitis and Impetigo, and two classes of fungi skin diseases namely Ringworm and Sporotrichosis. Healthy Skin images are the fifth class in the dataset. The Convolutional Neural Network (CNN) is known to be the most suitable deep learning algorithm for image processing applications. The power of the transfer learning technique is utilized by applying the fine-tuned version of VGG16 architecture. A web application is developed using Streamlit to provide a smooth user interface for displaying the predicted result of a skin image. The web app facilitates a user to either upload an image or captures a real-time image. The developed solution achieves an accuracy of 86% and an F1-score of 85%. Such a deep learning model will assist dermatologists in decision making, reduce the time required for diagnosis, and also minimizes the cost involved by a significant amount.


# 1. Introduction

The human skin is the largest and outermost organ of the human body. In addition to water and proteins, the skin also contains fats and minerals. The skin primarily protects the human body from harmful substances existing outside the body and also prevents the outflow of various nutrients present inside the human body. Skin is a very sensitive part of the body. Various internal and external factors harm the functionality of the skin which leads to various skin diseases. The epidermis, dermis, and hypodermis make up the three main layers of the skin, and all three layers are susceptible to skin diseases that can be life-threatening in the long run.

The problem of skin diseases is a global concern that affects people of all ages. Symptoms such as burning, redness, itching, and swelling can be caused by anything that harms the skin. Allergic reactions can irritate the genetic structure of the skin which causes rashes and several skin conditions that can negatively impact the appearance of the skin. Skin diseases are chronic and may grow into malignant tissues. Hence, to minimize the major damage, it should be identified and treated immediately.

Several skin diseases in humans may not exhibit symptoms at an early stage. They should be diagnosed and treated before they could cause serious problems. Dermatology is an area of medicine that deals with issues related to hair, nails, and skin. Traditionally, the diagnosis of a skin disease usually involves clinical screening and then a dermoscopic analysis. Dermatologists often have to perform laboratory

tests before concluding the type and name of the skin disease. Despite the advancements in medical equipment giving accurate results, the cost of such a diagnosis seems expensive and is a tedious task to some extent. Thus, there is a need for providing a quick and cost-effective solution. Such a solution can be developed by using deep learning techniques.

## 2. Literature Survey

The most common form of the disease in humans is a skin disease. The causes of skin diseases include fungal infections, bacterial infections, viruses, etc. Dermatologists play a crucial role in the traditional method of skin disease identification. A dermatologist observes a patient in the first instance to gather the skin condition based on the knowledge and experience gained. This is followed by a skin imaging process known as dermoscopy for observing the skin structure.

The appearance of skin diseases in an image plays a crucial role in diagnosing the type of the disease. Considering the difficulties in diagnosing skin diseases with traditional methods, computer-aided methods are being used in medicine [1].

Traditional approaches to skin disease diagnosis have been replaced with machine learning methods to overcome their limitations. It requires the use of manual extractors for the extraction of features of skin diseases and then using machine learning algorithms for classification. Further, since this process is performed manually, it requires professional medical knowledge as well as the ability to conduct deep exploratory data analysis to reduce dimensions that limit its capability when it comes to recognizing skin disease images [2]. The deep learning methods outperform the machine learning techniques for image recognition and classification since the former automates the feature engineering process.

A lot of researchers have been interested in using image recognition-based deep learning techniques to diagnose skin diseases due to the advancement of deep learning technology. In [3] authors have used a hybrid approach that involves deep feature fusion, and several SVM classifiers. It combines the probabilities from multiple classifiers to obtain the final classification. In the researchers' study, three pre-trained deep learning models were used as deep feature generators, including VGG16, AlexNet, and ResNet-18. The last fully connected layers of the pre-trained AlexNet and pre-trained VGG16 were used for extracting the features. Since ResNet-18 has only one fully connected layer, the features were extracted from the last convolution layer of this model. ROC curve was used as an accuracy metric. The combination of all networks resulted in 97.55% and 83.83% for classifying melanoma and seborrheic keratosis diseases respectively.

Reference [4] focuses on computer vision algorithms and various image processing algorithms for feature extraction, and ANNs (Artificial Neural Networks) are used to train and test the algorithms. It first preprocesses the images of the skin to extract specific features, then determines the disease type. Eight image processing algorithms were used. They were median filter, binary mask, smooth filter, sobel operator, gray image, histogram, sharpening filter, and YCbCr. Ten different features were used for

modeling the data. Using them a test accuracy of 90%, 85%, and 88% was achieved for supervised, unsupervised semi-supervised systems respectively.

For identifying histopathological characteristics of clinically evaluated samples, the authors of [5] used computer vision and machine-learning methods. They explored 6 diseases namely pityriasis rosea, chronic dermatitis, pityriasis rubra pilaris, seborrheic dermatitis, psoriasis, and lichen planus. They used decision trees, KNN for classification, and the ANN model which gave an accuracy of 95%.

Reference [6] makes use of AlexNet for feature extraction and SVM classifiers for classification. ANN and CNN (Convolutional Neural Networks) are both widely used for diagnosing skin diseases. Skin disease diagnosis using the CNN approach showed that the results are promising [7].

Reference [8] provides a comparative analysis of different CNN architectures to detect skin bacterial infections. ResNet50, Xception, InceptionV3, VGG16, and VGG19 architectures are studied. They also initially used the K-Fold cross-validation technique. Their network involved Adam optimizer and binary cross-entropy which is a loss function. The dropout rate was set to 25%. An accuracy of 91.303% and 84.545% was achieved using VGG16 and VGG19 architectures.

Reference [9] presents a web application system for classifying skin diseases using CNN architecture and the TensorFlow framework. Acne vulgaris and atopic dermatitis were some of the diseases considered in their study.

Reference [10] provides a workflow involving computer vision techniques that could be utilized for both mobile and web application systems.

Reference [11] proposed a method of clustering images that can be applied to the classification task. For detecting the key features in the image, SIFT method was used. Then SVM classifier and, segmentation technique was adopted resulting in a precision of 82% and an accuracy of 84%.

Reference [12] describes the formulas for image segmentation and feature extraction of the image. For Feature Extraction various parameters are calculated such as mean, Variance, Energy, and Entropy from the image.

Reference [13] combines InceptionResnetV2, MobileNet, and InceptionV3. The layers of these architectures are modified to suit their dataset for skin disease classification. The maximum voting rule was used for attaining the classification result. With an accuracy level of 88%, the authors were able to predict 20 diseases.

Reference [14] performs a multi-model approach for skin lesion classification. The authors have used the transfer learning technique to utilize both ResNet-101 and ResNet-50 for feature extraction. For reducing the dimensions of the feature an algorithm called KcPCA (Kurtosis controlled Principle Component Analysis) was developed. This process ensured the selection of optimal features. For classification, SVM with a radial basis function is utilized.

Reference [15] describes the Generative Adversarial Network (GAN) which can automatically discover and learn the patterns in input data. This way that the model can generate new samples from the original samples. GANs architecture consisting of CNNs can generate simulated data that is nearly a match for real data distribution. Features are extracted using MobileNet, GoogLeNet, ResNet, and DenseNet but DenseNet-121 and DenseNet-169 have produced the best accuracy of 94.25% and 93.67%.

Two stages were formed involving computer vision and machine learning techniques. This approach was performed on histopathological characteristics that were clinically validated. In the first step, the skin disease image is preprocessed and then the features are extracted. In the second step, the collected histopathological attributes are examined using machine learning algorithms. The experimental results for this proposed program are based on PyCharm-based Python scripts. The user interface web app was rendered in the project, where the image is uploaded and results are displayed [16].

Reference [17] utilizes Deep Convolutional Neural Networks (DCNN) for image classification and feature extraction processes. The layers consisted of three convolutional and pooling layers, followed by a fully connected layer for feature extraction. Their work can be viewed as two sub-tasks. The first task involves the extraction of features from DCNN, and the extracted information is fed into a neural network of two layers for classification. The second task involves classification by SVM. The results of their experiment show that the SVM classification on DCNN features is better than the performance of the neural network alone for the classification task. From their experimentation, it is understood that having several feature maps helps in discovering different patterns from the input image at several locations. The error on testing data starts to increase slowly once the number of feature maps and convolutional layers exceeds a certain limit. The training error increases after several iterations with a slightly high learning rate.

For image classification, [18] proposes a method for creating automatic CNN architectures using genetic algorithms (CNN-GA). Because of their algorithm, users with little to no experience in tuning CNN architectures can still find an appropriate CNN architecture from the provided images. For producing deeper CNNs, the authors have developed an encoding technique for the genetic algorithms that can encode CNNs having any depth. Two components for CNN-GA are designed to expedite analysis and save a lot of computing resources. Considering the computing resources consumed, the number of parameters required and the classification accuracy, they found that the proposed algorithm outperforms the existing automatic CNN architectures.

Reference [19] developed a new deep learning model that is named Convolutional eXtreme Gradient Boosting (ConvXGB) for classification problems. This work was carried out on several image datasets and general datasets available in the UCI repository. From an initial point of view, they considered that a single model didn't provide the amount of accuracy required for various models. As the name suggests, their model combines the Convolutional Neural Networks and the XGBoost classifier. ConvXGB also includes a module for preprocessing data. The ConvXGB model learns the input features using several convolutional layers, and XGBoost is used for class label prediction in the final layer. It lacks a pooling layer or a Fully Connected layer as opposed to the traditional CNN. This way the number of parameters is

simplified in this model because the weight values do not have to be readjusted in the backpropagation cycle. As a result of the experiment, their ConvXGB model generally outperformed CNN and XGBoost alone for all data sets and, in some cases, by a significant margin.

Reference [20] utilizes the transfer learning technique by using the VGG-16 architecture. It is capable of extracting 1000 features from the input image. Apart from SVM, and decision tree algorithms for classification, the K-Nearest Neighbor and linear discriminate analysis were performed. ICIS public datasets were used for this study that consists of malignant and benign types of disease. The use of transfer learning techniques is found to be advantageous as it increases the accuracy level when compared with other pure deep learning models. Using the VGG-16 CNN model in conjunction with the K-Nearest Neighbor algorithm was found to produce the best results in the experiment. On the other hand, complex models like ensemble learning using boosted trees performed at or below 50%. By observing these results, the authors concluded that the dataset used is best suited only for linear binary classifiers.

Reference [21] uses a triplet loss function for skin disease classification using CNN. The authors fine-tuned the layers of InceptionResNet-V2 and ResNet152 to suit the classification needs. In the first step, 128-D embedded features from training samples are extracted into the Euclidean space. In the next step, the learned embeddings are used to compute the L-2 distances between the images. The L2 distances computed among the images are used for classifying skin diseases. This experiment analyzed four types of skin diseases: dark circles, acne, spots, and blackheads. The method was evaluated using 12000 input images for training and 2000 for testing. Also, 10% of images from the training set were used for validation.

From the literature survey, it is inferred that deep learning techniques have a greater impact on the diagnosis of skin diseases. In addition, CNN architectures are popular for the task of image processing. Also, the transfer learning technique allows using existing architectures of CNN for fine-tuning them or directly applying them using their previously learned weights for the classification task.

# 3. Methodology

The following techniques were implemented:

- Method 1: A New CNN architecture
  A CNN network was designed from scratch. Various combinations of convolution and pooling layers were added to the network.
- Method 2: Using Machine Learning classifiers with CNN architectures
  Two pre-existing CNN architectures were utilized for feature extraction. After the feature extraction, classifiers like SVM, Random Forest, and XGBoost were added.
- Method 3: Combining features obtained from different CNN architectures
  Two CNN architectures were combined for feature extraction and then a classifier was used.

- Method 4: Retraining a few layers of a pre-trained architecture
  Finally, as part of the Transfer Learning technique, various pre-existing networks were fine-tuned to suit the needs of the dataset.

Among these methods, method 4 performed better.

# 3.1. Transfer Learning

Transfer Learning is the process of transferring the feature maps learned by one neural network that was trained on a task containing enough data, to a new task where the amount of data is insufficient for training. The neural network from which feature maps are utilized to solve another task is known as a pre-trained network. It means that transfer learning involves using neural networks trained on one task as the initial state for working on a related task. The transfer learning should be applied to the task in consideration only if both the tasks are similar to each other. Since the collected dataset has insufficient images for training, the transfer learning approach is best suited considering it reduces the amount of time required for training.

For an image classification task, Transfer learning offers flexibility in the following four ways:

- The pre-trained network can be applied to a classification task without making any changes to the pre-trained network.
- The pre-trained network can be used only for the process of feature extraction.
- Some layers of the pre-trained model can be integrated into a new neural network but the layers of the pre-trained model can be frozen during the training process.
- Finally, some portion of the pre-trained network can be integrated into a new neural network, and then those layers can be retrained using the dataset.

VGG16 is the pre-trained network that is considered for solving the identified problem statement. The VGG16 model was trained on a subset of the ImageNet database. Originally, ImageNet was a project designed to classify and label images into more than 20,000 distinct categories for computer vision research. For CNN and deep learning, ImageNet refers to the Image Large Scale Visual Recognition Challenge (ILSVRC). The objective of ILSRVC is to build a model that can accurately classify input images into 1,000 categories. ILSVRC is a subset of ImageNet. From each of the 1000 categories, ILSVRC contains around 1000 images. These 1000 categories represent objects that we encounter in daily life such as vehicles, household objects, cats, dogs, etc.

VGG16 consists of 21 layers but only 16 layers of those are trainable. The architecture includes 13 convolutional layers, 5 max-pooling layers, and 3 fully connected layers. Several small filters are used throughout the network with max-pooling operations being performed periodically. Instead of using large filters, VGG16 architecture considers smaller filters with more depth in the network. These small-sized filters have a size of 3x3 and 1x1 with a stride of one. Max pooling layers are used with a size of 2x2 and

stride size 2. The model starts with 64 filters and gradually expands to 128 and 256 as the depth of the model is increased, and finally up to 512 filters when the process of feature extraction ends. Figure 1 represents the layers of VGG16 architecture.

The task of identifying and classifying skin diseases involves medical images. The collected dataset consists of 5 classes namely, Impetigo (Bacterial), Cellulitis (Bacterial), Sporotrichosis (Fungal), Ringworm (Fungal), and Healthy Skin. This contains nearly 500 images where the frequency of each class is unequal.

These images are quite different from the images present in the ImageNet database. Accordingly, VGG16 is neither appropriate for entirely using it for classification nor using the same weights for the feature extraction process. Instead, the layers of the VGG16 architecture are fine-tuned by freezing the initial convolutional layers that extract the features using the pre-trained weights and then by retraining the other convolutional layers with the skin disease images from the dataset.

## 4. Preprocessing

All the images in the dataset need to undergo some kind of pre-processing step to ensure that a CNN model obtains the images in the required format. This process simplifies the feature extraction ability of the neural network model by eliminating the irrelevant information in the image. The pre-processing methods applied to the dataset are Image Resizing, Label Encoding, One-hot encoding, Normalization, Class Weights, and Image Augmentation.

## 4.1. Image Resizing

A CNN model expects all the input images to be of the same dimensions. All the images of the dataset are resized to 256 x 256.

## 4.2. Label Encoding and One-hot Coding

Label encoding refers to the process of converting class labels into a numerical format so that they are machine-readable. The 5 class labels of the dataset are Cellulitis_Bacteria, Healthy_Skin, Impetigo_Bacteria, Ringworm_Fungus, and Sporotrichosis_Fungus. These class labels are converted to the integers 0,1,2,3, and 4 respectively. Each of these encoded labels is converted as individual columns using the one-hot encoding technique.

## 4.3. Normalization

Normalization places the pixel values of the images into a common statistical distribution. For the image dataset in consideration, each pixel of an image is divided by 255 which places all the pixel values in the common range of [0,1]. This is an important step that ensures faster convergence while training the neural network.

## 4.4 Balancing Dataset

Each class in the collected dataset does not have an equal number of images. Balancing Dataset ensures the CNN model is not biased towards the majority class of the dataset. For the image dataset in consideration, the Class Weight method is performed. If class_weights is set to 'balanced', the model automatically adjusts the class weights inversely proportional to the frequency of each class.

## 4.5 Image Augmentation

Image Augmentation creates slightly modified copies of the existing images in the dataset. This technique is generally performed when there are insufficient images for training the CNN model. It reduces overfitting and increases the ability of the CNN model to generalize while extracting the features from the images. The augmentation techniques such as width shift, height shift, zooming, and vertical, and horizontal flips are performed. The augmented images are generated during the model training using the Keras ImageDataGenerator. This prevents the storage of augmented images which saves a lot of storage space. The training images are augmented while the testing images are not augmented.

# 5. Algorithm

The proposed solution is based on a deep learning algorithm. CNN is a type of deep learning algorithm that consists of convolutional layers, pooling layers, and fully connected layers. CNN is mainly used for image classification tasks.

An Image consists of numbers known as pixel values. The filters (kernels) consist of numbers known as weights which are randomly initialized values. VGG16 is a type of CNN architecture developed by Andrew Zisserman and Karen Simonyan. The fine-tuned version of the VGG16 architecture has been utilized for the proposed solution. This approach to solving a task is termed transfer learning

Input: The pre-processed training images are fed to the first convolutional layer. All the input images are of size 256 x 256.

Forward Pass: The input images go through various convolutional layers and pooling layers. At each convolutional layer, an input image is convoluted on the filter. Pooling layers can be added periodically after convolutional layers. After going through all the CNN layers, patterns and features are detected that result in what is known as feature maps.

The feature maps are utilized by the model to predict the class of the images. The predicted output is compared with the expected output to compute the error in the prediction.

Backward Pass: During the backward pass, the backpropagation algorithm updates the value of the filters at each layer based on the prediction error. The input images undergo the process of the forward pass and backward pass depending on the number of epochs to find out the optimal values for the filters that result in the least error while prediction. This is known as the weight updating process.

Output: The pre-processed test images are fed the trained model for prediction. The trained model predicts the class of the images based on the learned feature maps.

# 6. Model Development

The implementation is carried out using the Python Programming language. The model is developed using the Keras framework. The dataset is loaded by mounting the Google Drive to the Google Colab notebook. All the images are resized and class labels are encoded to integers. The dataset is split such that 80% of the dataset is used for training while 20% of the data is used for validation and testing.

A stratified split is performed that forces the distribution of the classes among different splits to be the same. The training and testing images are normalized and then the class weights are computed. The class labels of the training and testing images undergo one hot encoding before they are fed to the VGG16 model.

The VGG16 model is incorporated by excluding the fully-connected layers. Except for the last 5 layers of the VGG16 model, all other layers are freezed to retain the pre-trained weights. A dense layer with 5 neurons is added to the architecture with softmax as the activation function. The last 5 layers of the VGG16 model are retrained using the training images to find out the optimal weights for those layers.

The training images are augmented while training the VGG16 model. The model is compiled by using an Adam optimizer with learning set to 0.00001 and categorical cross-entropy as the loss function.

A fine-tuned version of the VGG16 model is obtained by training the unfreezed layers for 150 epochs with batch size set to 32.

The summary of the fine-tuned VGG16 model is shown in Fig. 2.

# 7. Results

The developed model is tested with the testing set images that consisted of 97 images. The model achieved a test accuracy of 86% and an F1-score of 85%.

Figure 3. represents the loss that occurred during the training and validation whereas Fig. 4. represents the accuracy obtained during the training and validation.

Figure 5. represents the confusion matrix generated for the test dataset. In a classification problem, a confusion matrix summarizes the results of the prediction. The confusion matrix indicates how a classification model is confused when it makes predictions. Not only does it give insight into the errors occurring in the classifier, but it also tells us what types of errors are occurring.

Figure 6. represents the classification report generated for the test dataset. The classification report serves as a performance evaluation metric. An evaluation of the trained classification model is done by

measuring its precision, recall, F1 Score, and support. The ratio of true positives to the sum of true and false positives is called precision. The ratio of true positives to the sum of true positives and false negatives is called recall. F1-score is the measure of the weighted harmonic mean of recall and precision. It is maximum when precision is equal to recall. The model is expected to perform better as the value of the F1-score gets closer to 1.0.

Figure 7. Represents the actual disease and predicted disease for some of the test images. In addition, the confidence of prediction is also displayed which represents the probability of prediction.

# 8. Web Application

A web application is developed using Streamlit, an open-source application framework for Data Science projects. The developed application provides a smooth user interface to display the predictions of an input image. The name of the web app is DermoSpect. On the Home page, a short description of the problem statement is provided. The application allows the prediction of skin diseases using two methods. The first method is by using a real-time camera. The image of the skin is captured on spot by the user. The second method is by uploading an image of the skin saved in the user's system. In any of the methods, an image given as input by the user will be fed to the trained VGG16 model after it undergoes all the pre-processing steps. The model outputs the predicted disease type, predicted disease name, and the probability of prediction. This output is displayed on the web application to the user.

The overall procedure of the work is represented in Fig. 8. The input images first undergo the pre-processing steps. Once the images are pre-processed, the convolutional layers of the VGG16 model extract the features. The dense layer of the VGG16 model outputs the prediction. Finally, the prediction is displayed on the web application. The layout of the web application is represented in Fig. 9.

The snapshots from the web application are represented by the following figures.

# 9. Conclusion

A solution is presented with non-dermoscopic images as the input to the developed model. The power of transfer learning is utilized by developing a fine-tuned version of the VGG16 model which is 86% accurate. The developed web application provides a smooth user interface that displays the predictions with the click of a button. The web application facilitates uploading an image and capturing an image from the web camera as well. Apart from displaying the predicted disease type and name, the model also outputs the probability of the prediction being true.

The developed model will assist a dermatologist in decision making, reduces the time required for diagnosis, and minimizes the cost involved by a significant amount.

# Declarations

**Acknowledgment**

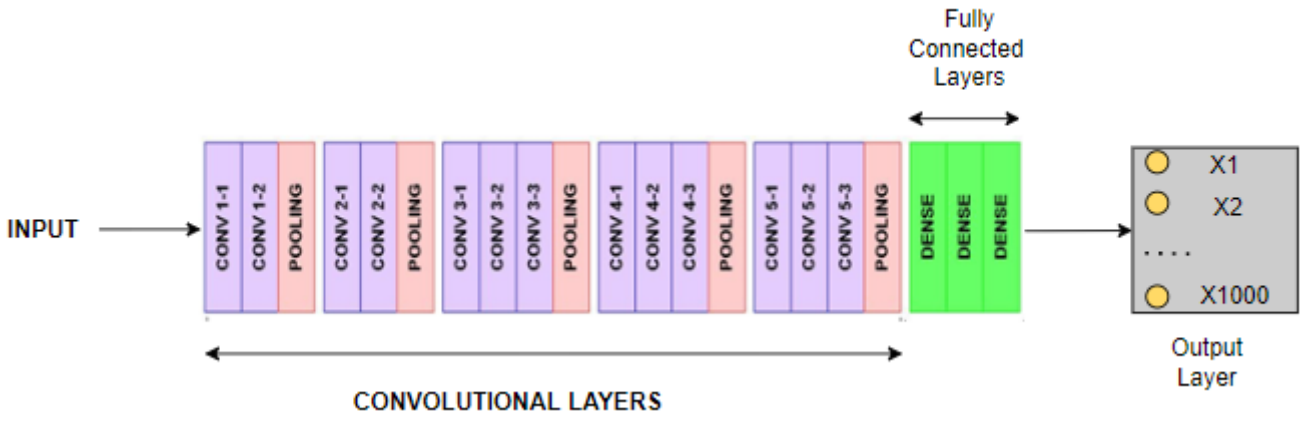
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# Figures



## Figure 1

Layers of VGG16 architecture

| Layer (type) | Output Shape | Param # |
|---|---|---|
| input_1 (InputLayer) | [(None, 256, 256, 3)] | 0 |
| block1_conv1 (Conv2D) | (None, 256, 256, 64) | 1792 |
| block1_conv2 (Conv2D) | (None, 256, 256, 64) | 36928 |
| block1_pool (MaxPooling2D) | (None, 128, 128, 64) | 0 |
| block2_conv1 (Conv2D) | (None, 128, 128, 128) | 73856 |
| block2_conv2 (Conv2D) | (None, 128, 128, 128) | 147584 |
| block2_pool (MaxPooling2D) | (None, 64, 64, 128) | 0 |
| block3_conv1 (Conv2D) | (None, 64, 64, 256) | 295168 |
| block3_conv2 (Conv2D) | (None, 64, 64, 256) | 590080 |
| block3_conv3 (Conv2D) | (None, 64, 64, 256) | 590080 |
| block3_pool (MaxPooling2D) | (None, 32, 32, 256) | 0 |
| block4_conv1 (Conv2D) | (None, 32, 32, 512) | 1180160 |
| block4_conv2 (Conv2D) | (None, 32, 32, 512) | 2359808 |
| block4_conv3 (Conv2D) | (None, 32, 32, 512) | 2359808 |
| block4_pool (MaxPooling2D) | (None, 16, 16, 512) | 0 |
| block5_conv1 (Conv2D) | (None, 16, 16, 512) | 2359808 |
| block5_conv2 (Conv2D) | (None, 16, 16, 512) | 2359808 |
| block5_conv3 (Conv2D) | (None, 16, 16, 512) | 2359808 |
| block5_pool (MaxPooling2D) | (None, 8, 8, 512) | 0 |
| global_average_pooling2d (GlobalAveragePooling2D) | (None, 512) | 0 |
| softmax (Dense) | (None, 5) | 2565 |

Total params: 14,717,253
Trainable params: 7,081,989
Non-trainable params: 7,635,264

## Figure 2

Summary of fine-tuned VGG16 model

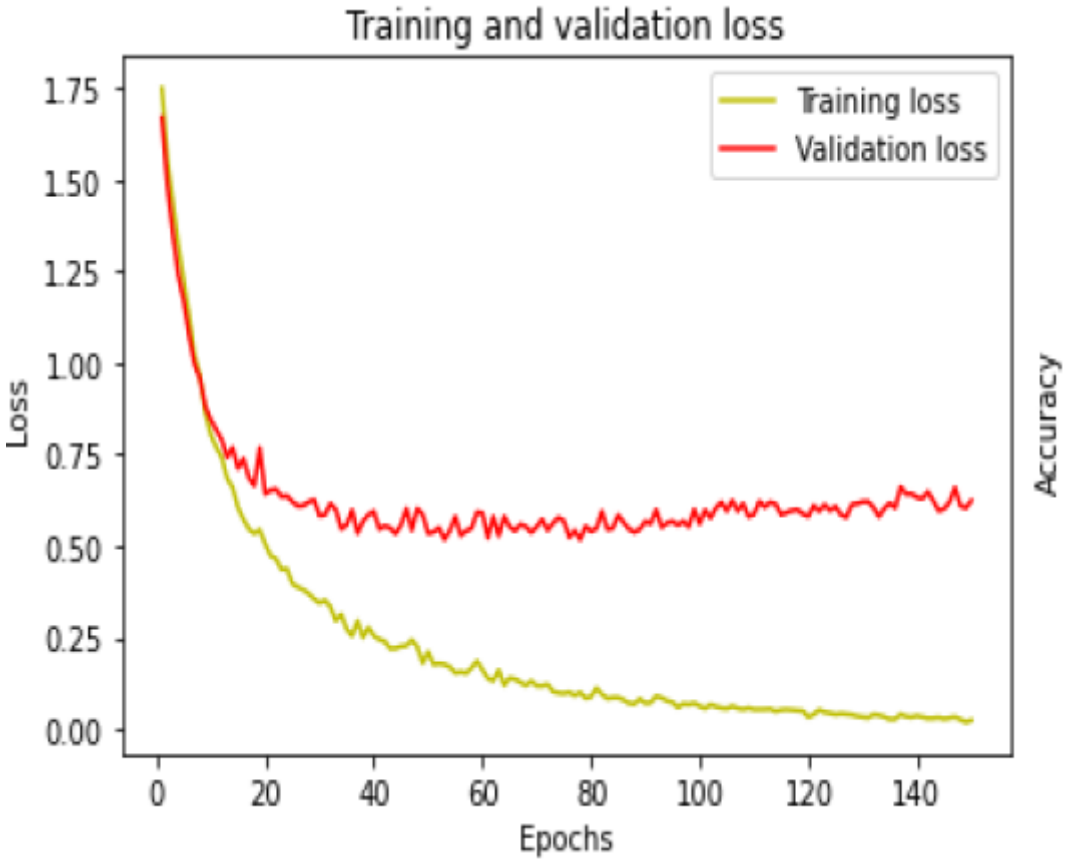



**Figure 3**

Training and Validation loss




**Figure 4**

Training and Validation accuracy

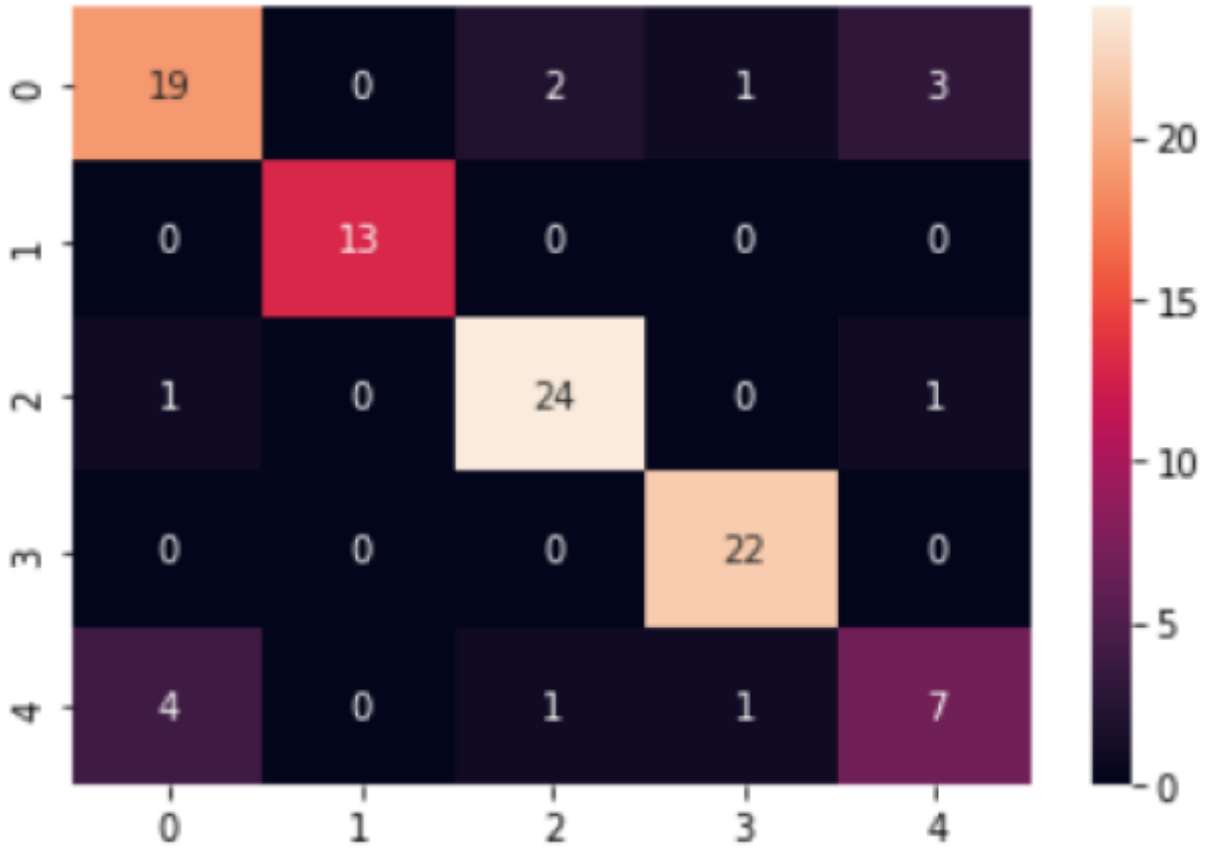



Fig 6.2 Confusion Matrix

**Figure 5**

Confusion Matrix

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.79 | 0.76 | 0.78 | 25 |
| 1 | 1.00 | 1.00 | 1.00 | 13 |
| 2 | 0.89 | 0.92 | 0.91 | 26 |
| 3 | 0.92 | 1.00 | 0.96 | 22 |
| 4 | 0.64 | 0.54 | 0.58 | 13 |
| accuracy | | | 0.86 | 99 |
| macro avg | 0.85 | 0.84 | 0.84 | 99 |
| weighted avg | 0.85 | 0.86 | 0.85 | 99 |

**Figure 6**

Classification Report

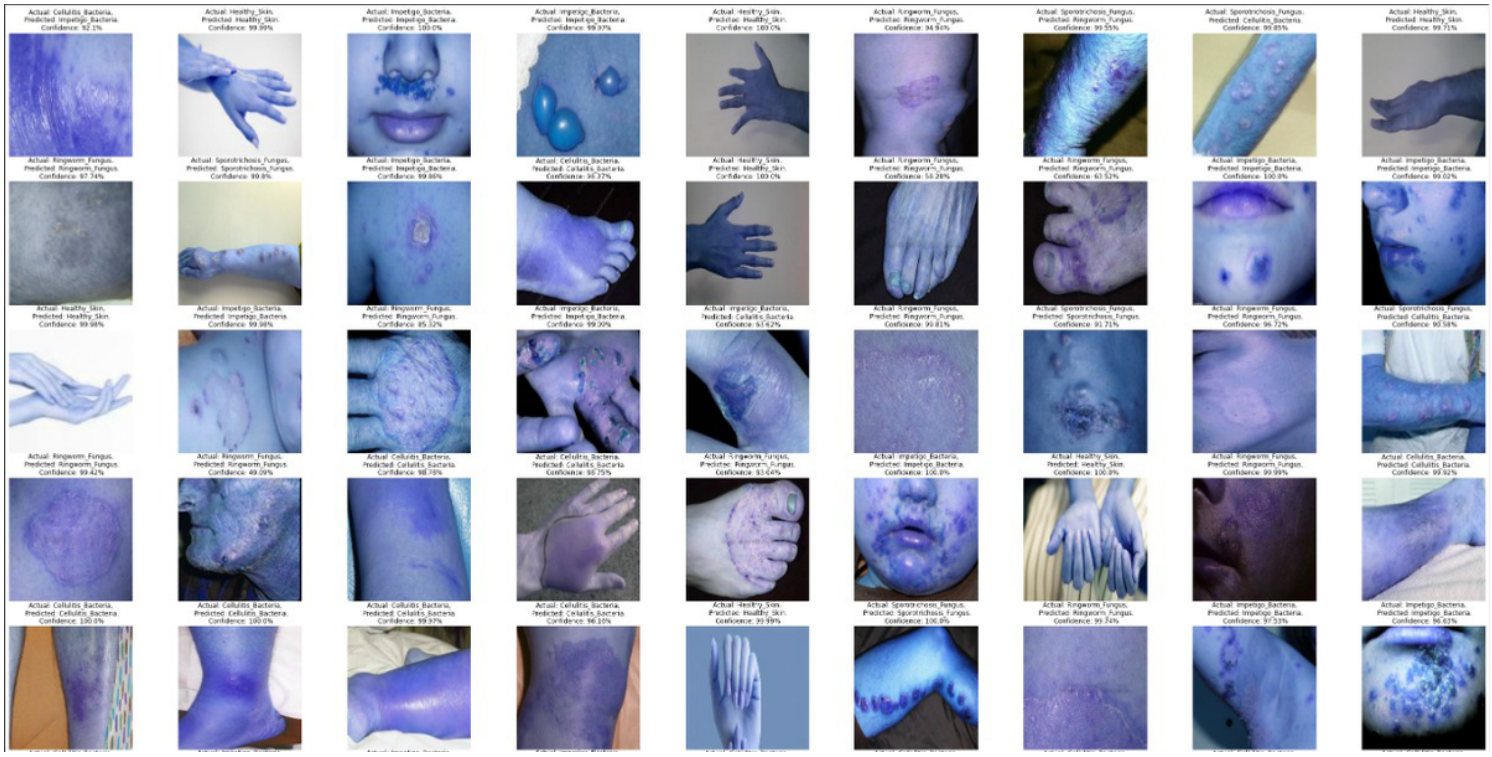


**Figure 7**

Output obtained from the test images

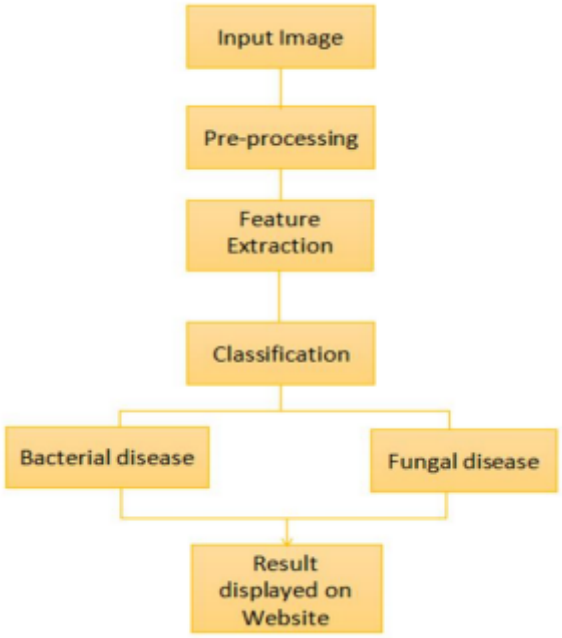



**Figure 8**

Overall Procedure

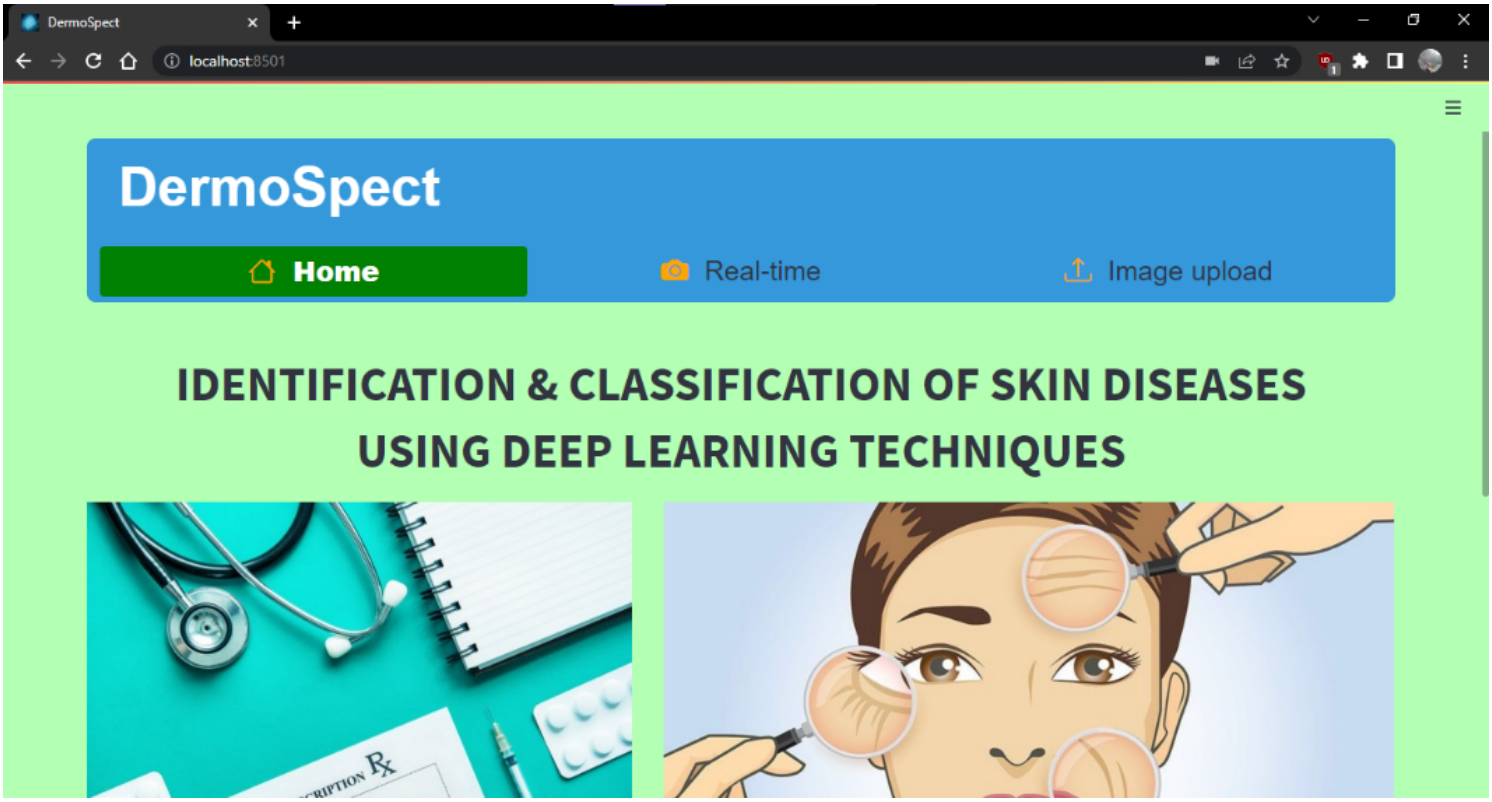



**Figure 9**

Layout of the web application

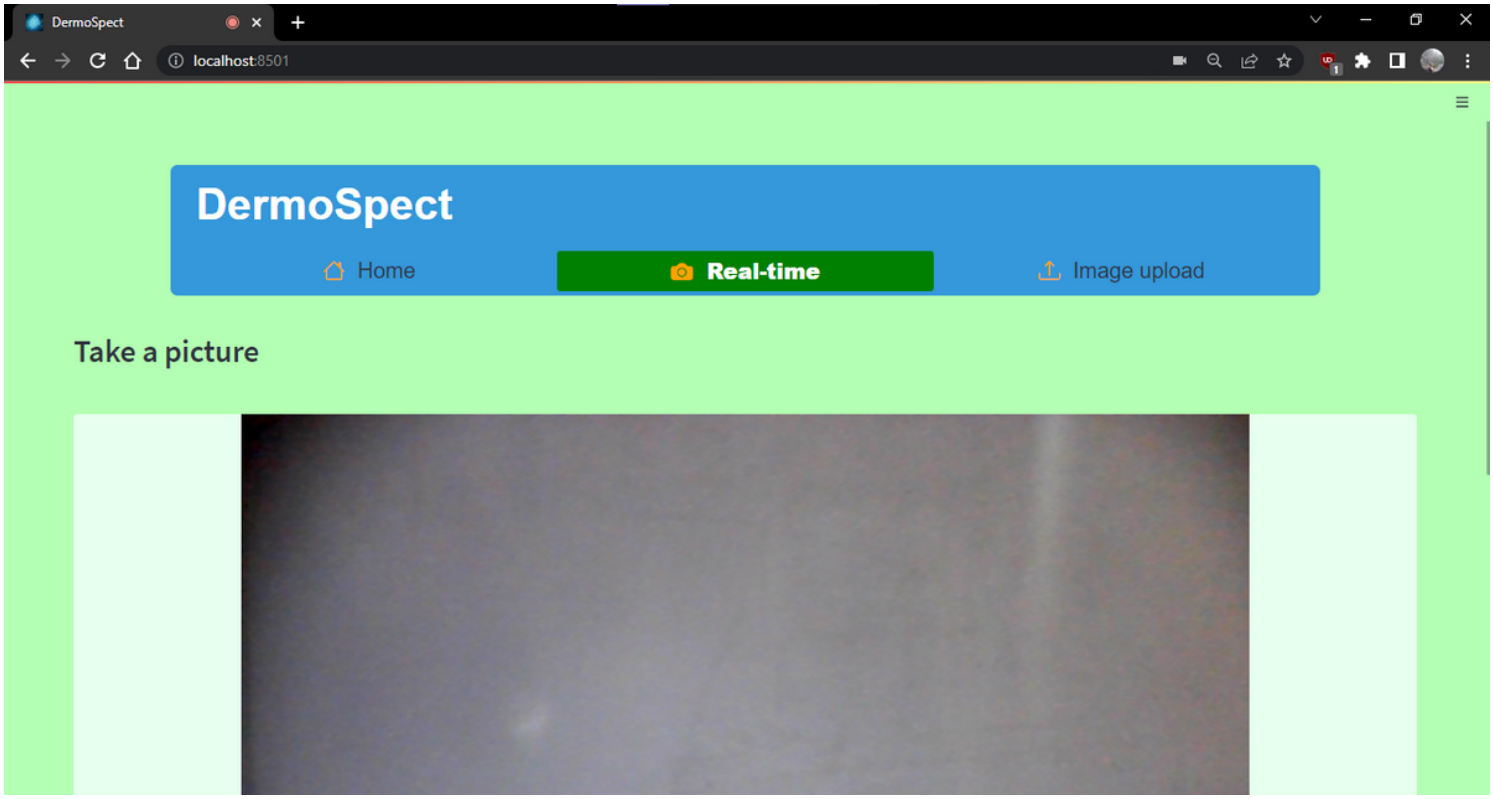



Figure 10

Real-time capturing of the image with a web camera

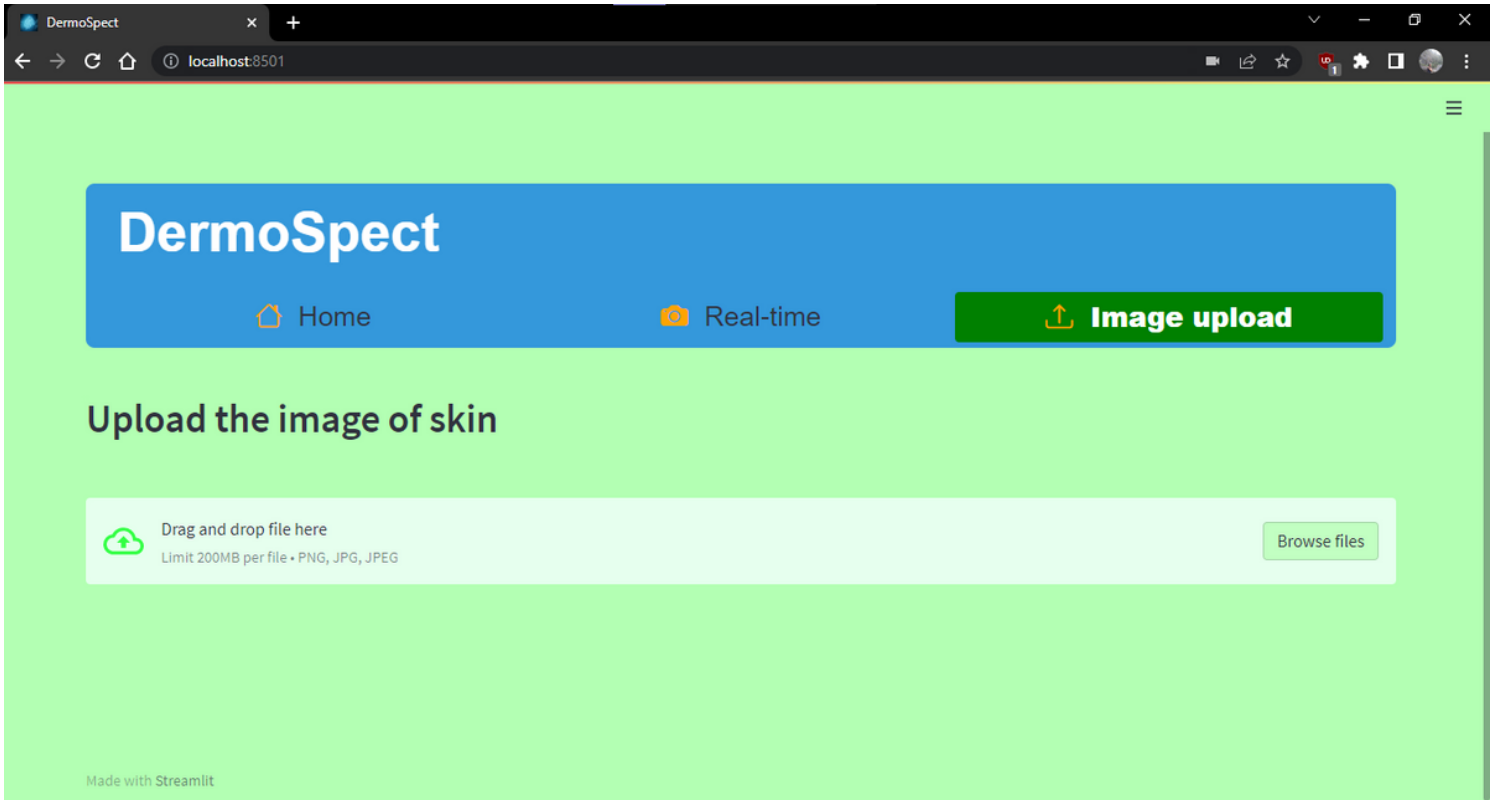


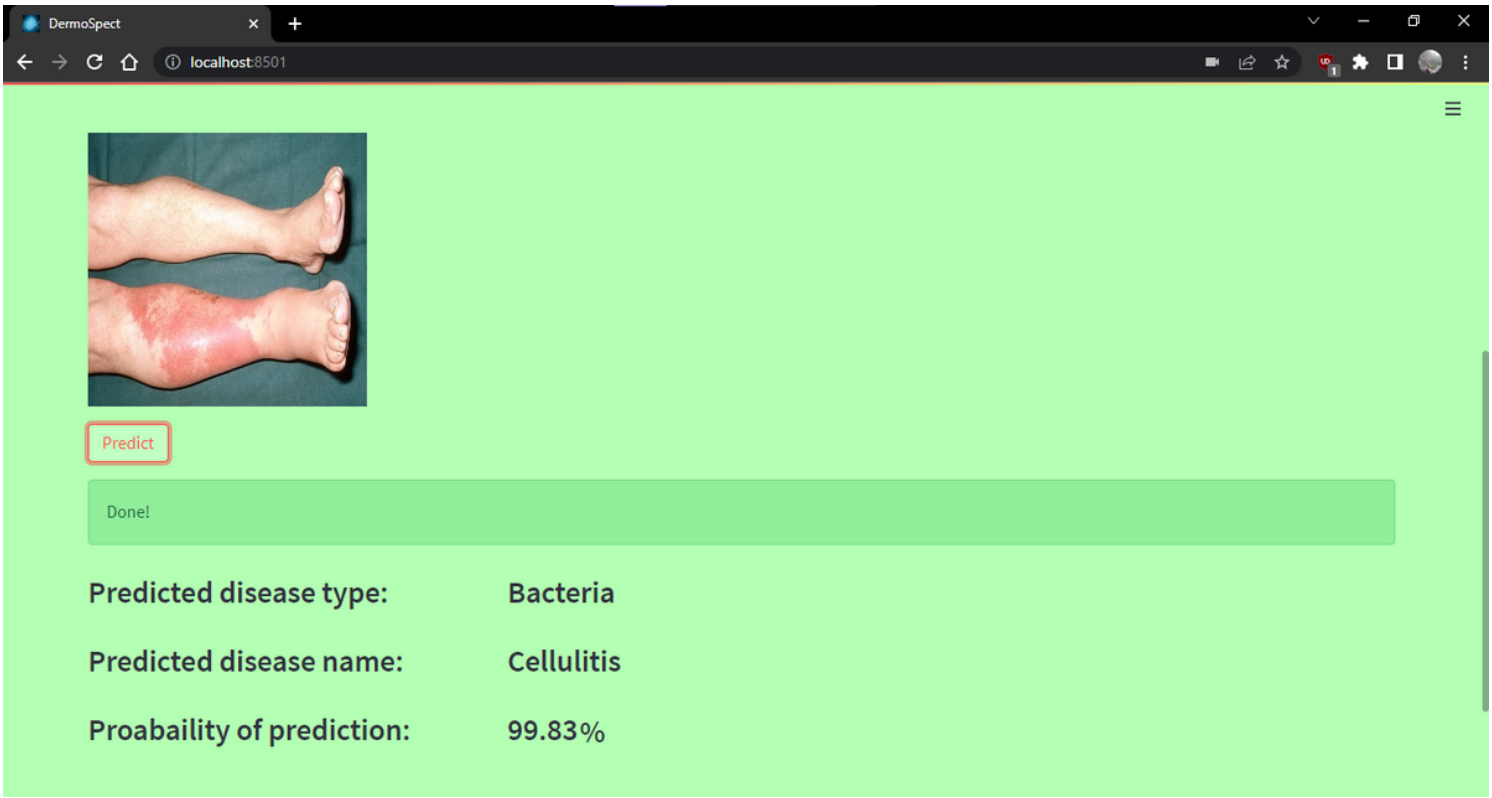

Figure 11

Uploading Images from the system




**Figure 12**

Healthy Skin Prediction



## Figure 13

Prediction of disease uploaded from the system (Cellulitis disease)